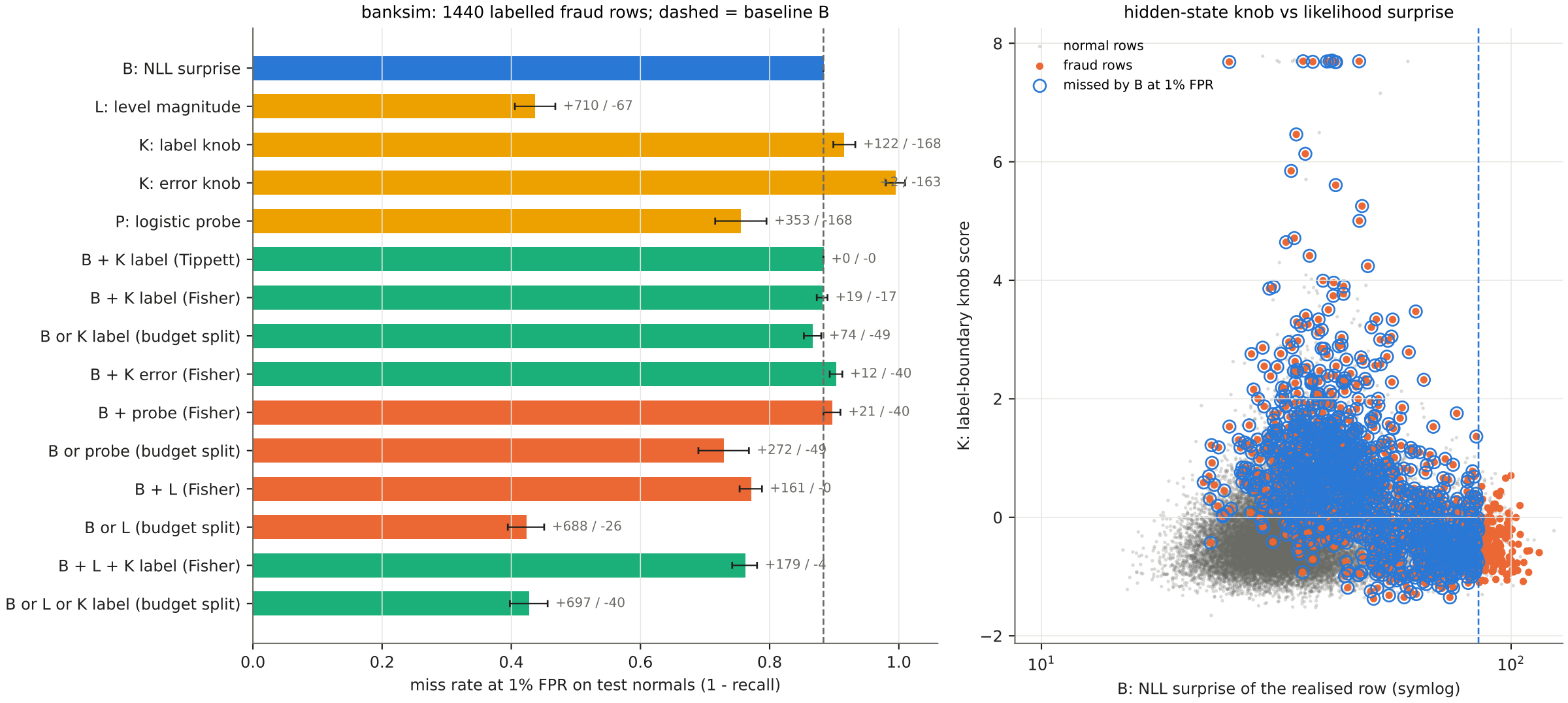

banksim: 1440 labelled fraud rows; dashed = baseline B
B: NLL surprise
L: level magnitude
+710 / -67
K: label knob
+122 / -168
K: error knob
+2 / -163
P: logistic probe
+353 / -168
B + K label (Tippett)
+0 / -0
B + K label (Fisher)
+19 / -17
B or K label (budget split)
+74 / -49
B + K error (Fisher)
+12 / -40
B + probe (Fisher)
+21 / -40
B or probe (budget split)
+272 / -49
B + L (Fisher)
+161 / -0
B or L (budget split)
+688 / -26
B + L + K label (Fisher)
+179 / -4
B or L or K label (budget split)
+697 / -40
0.0
0.2
0.4
0.6
0.8
1.0
miss rate at 1% FPR on test normals (1 - recall)
hidden-state knob vs likelihood surprise
normal rows
fraud rows
missed by B at 1% FPR
K: label-boundary knob score
8
6
4
2
0
−2
$10^1$
$10^2$
B: NLL surprise of the realised row (symlog)